Assessment

Assessment process:

Is a systematic method by which HCP :plans and provides care for patients.

This involves a problem-solving approach that enables the HCP to identify patient problems and potential at-risk needs (problems) and to plan, deliver, and evaluate the care in an orderly, scientific manner.

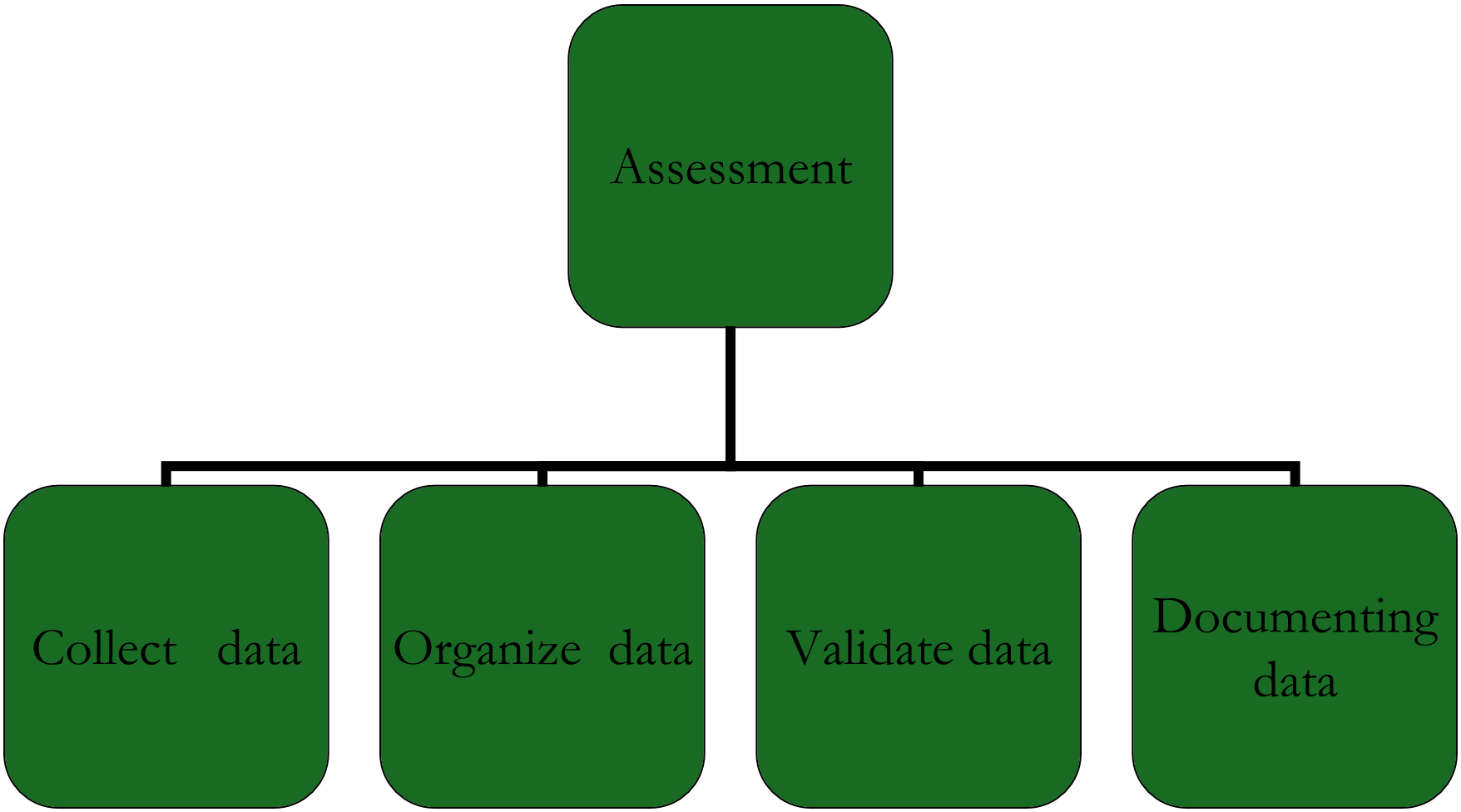
Assessment

Is the systematic and continuous:

- Collection
- Organization
    - Validation
        - Documentation of data.



The Process


Assessment
Collect data
Organize data
Validate data
Documenting data

The Process

- The HCP gathers information to identify the health status of the patient.
- Assessments are made initially and continuously throughout patient care.
- The remaining phases of the care process depend on the validity and completeness of the initial data collection.

Purposes of assessment

To establish Database: all the information about a client: it includes:

- Health history
- Physical examination
    - The physician's history
        - Results of laboratory and diagnostic tests

PURPOSE

Assessment is part of each activity performed with the patient.

The purposes is

1. To validate a diagnosis
2. To provide basis for effective care.
3. It helps in effective decision making
4. Basis for accurate diagnosis
5. It promote holistic care
6. To provide effective and innovative care
7. To collecting data for scientific research
8. To evaluation of nursing care and other professionals care

Types of Assessment

Assessment

Initial Assessment

Focus Assessment

Time-lapsed Assessment

Emergency Assessment

Initial comprehensive assessment

An initial assessment, also called an admission assessment, is performed when the client enters a health care from a health care agency. The purposes are to evaluate the client's health status, to identify functional health patterns that are problematic, and to provide an in-depth, comprehensive database, which is critical for evaluating changes in the client's health status in subsequent assessments.

Problem-focused assessment

A problem focus assessment collects data about a problem that has already been identified. This type of assessment has a narrower scope and a shorter time frame than the initial assessment. In focus assessments, nurse determine whether the problems still exists and whether the status of the problem has changed (i.e. improved, worsened, or resolved). This assessment also includes the appraisal of any new, overlooked, or misdiagnosed problems. In intensive care units, may perform focus assessment every few minute.

Emergency assessment

Emergency assessment takes place in life-threatening situations in which the preservation of life is the top priority. Time is of the essence rapid identification of and intervention for the client's health problems. Often the client's difficulties involve airway, breathing and circulatory problems (the ABCs). Abrupt changes in self-concept (suicidal thoughts) or roles or relationships (social conflict leading to violent acts) can also initiate an emergency. Emergency assessment focuses on few essential health patterns and is not comprehensive.



Time-lapsed assessment or Ongoing assessment

Time lapsed reassessment, another type of assessment, takes place after the initial assessment to evaluate any changes in the clients functional health. Nurses perform time-lapsed reassessment when substantial periods of time have elapsed between assessments (e.g., periodic output patient clinic visits, home health visits, health and development screenings)



Steps Of Assessment

A. Collection of data

a) Subjective data collection

b) Objective data collection

B. Validation of data

C. Organization of data

D. Recording/documentation of data

Collection of Data

- **gathering of information about the client**
- **includes physical, psychological, emotion, socio-cultural, spiritual factors that may affect client's health status**
- **includes past health history of client (allergies, past surgeries, chronic diseases, use of folk healing methods)**
- **includes current/present problems of client (pain, nausea, sleep pattern, religious practices, medication or treatment the client is taking now)**

Types of Data

When performing an assessment the HCP gathers subjective and objective data.



Subjective data (symptoms or covert data):

are the verbal statements provided by the Patient. Statements about nausea and descriptions of pain and fatigue are examples of subjective data.

Objective Data

Objective data (signs or overt data), are detectable by an observer or can be measured or tested against an accepted standard. They can be seen, heard, felt, or smelt, and they are obtained by observation or physical examination. For example: discoloration of the skin

Data Collection Methods

1. Observing: to observe is to gather data by using the senses.
2. Interviewing: an interview is a planned communication or conversation with a purpose.
3. Examining: Performance of a physical examination. The physical examination is often guided by data provided by the patient. A head-to-toe approach is frequently used to provide systematic approach that helps to avoid omitting important data

Physical assessment



Assessment Sequencing

- Head – to - Toe Assessment
- Body Systems Assessment

Head-to-toe Assessment

Physical Assessment using head toe approach

General
General health status
Vital signs and weight
Nutrional status

Mobility and self care
Observe posture
Assess gait and balance
Evaluate mobility
Activities of daily living

Head face and neck
Evaluate cognition
LOC
Orientation
Mood
Language and memory
Sensory function
Test vision
Inspect and examine ears
Test hearing
Cranial nerves
Inspect lymph nodes
Inspect neck veins

Chest
Inspect and palpate breast
Inspect and auscultate lungs
Auscultate heart

Abdomen
Inspect, auscultate, palpate four quadrants
Palpate and percuss liver, stomach, bladder
Bowel elimination
Urinary elimination

Cont.....

Skin, hair and nails

Inspect scalp, hair & nails
Evaluate skin turgor
Observe skin lesion
Assess wounds

Genitalia

Inspect female client
Inspect male client

Extremities

Palpate arterial pulses
Observe capillary refill
Evaluate edema
Assess joint mobility
Measure strength
Assess sensory function
Assess circulation, movement, & sensation
Deep tendon reflexes
Inspect skin and nails

Body System approach
Review Of Syste

General presentation of symptoms: Fever, chills, malaise, pain, sleep patterns, fatigability

Diet: Appetite, likes and dislikes, restrictions, written dairy of food intake

Skin, hair, and nails: rash or eruption, itching, color or texture change, excessive sweating, abnormal nail or hair growth

Musculoskeletal: Joint stiffness, pain, restricted motion, swelling, redness, heat, deformity

***Head and neck*:**

Eyes: visual acuity, blurring, diplopia, photophobia, pain, recent change in vision

Ears: Hearing loss, pain, discharge, tinnitus, vertigo

Nose: Sense of smell, frequency of colds, obstruction, epistaxis, sinus pain, or postnasal discharge

Throat and mouth: Hoarseness or change in voice, frequent sore throat, bleeding o swelling, of gums, recent tooth abscesses or extractions, soreness of tongue or mucosa.

Endocrine and genital reproductive: Thyroid enlargement or tenderness, heat or cold intolerance, unexplained weight change, polyuria, polydipsia, changes in distribution of facial hair; *Males:* Puberty onset, difficulty with erections, testicular pain, libido, infertility; *Females:* Menses {onset, regularity, duration and amount}, Dysmenorrhea, last menstrual period, frequency of intercourse, age at menopause, pregnancies {number, miscarriage, abortions} type of delivery, complications, use of contraceptives; breasts {pain, tenderness, discharge, lumps}

Chest and lungs: Pain related to respiration, dyspnea, cyanosis, wheezing, cough, sputum {character, and quantity}, exposure to tuberculosis (TB), last chest X-ray

Heart and blood vessels: Chest pain or distress, precipitating causes, timing and duration, relieving factors, dyspnea, orthopnea, edema, hypertension, exercise tolerance

Gastrointestinal: Appetite, digestion, food intolerance, dysphagia, heartburn, nausea or vomiting, bowel regularity, change in stool color, or contents, constipation or diarrhea, flatulence or hemorrhoids

Genitourinary: Dysuria, flank or suprapubic pain, urgency, frequency, nocturia, hematuria, polyuria, hesitancy, loss in force of stream, edema, sexually transmitted disease

Neurological: Syncope, seizures, weakness or paralysis, abnormalities of sensation or coordination, tremors, loss of memory

Psychiatric: Depression, mood changes, difficulty concentrating nervousness, tension, suicidal thoughts, irritability.

Pediatrics: along with systemic approach in case of pediatrics, measure anthropometric measurement and neuromuscular assessment.

Assessment techniques

- Inspection
- Palpation
- Percussion
- Auscultation
- The innovative Telemetry Monitoring System



Assessment techniques - Inspection

- Close and careful visualization of the person as a whole and of each body system
- *Ensure good lighting*
- *Perform at every encounter with your client*





Assessment techniques Palpation

- Temperature, Texture, Moisture
- Organ size and location
- Rigidity or spasticity
- Crepitation & Vibration
- Position & Size
- Presence of lumps or masses
- Tenderness, or pain

Palpation Techniques

- Light
- Deep



Assessment techniques Percussion

- assess underlying structures for **location, size, density of underlying tissue.**
- **Direct**
- **Indirect**
- **Blunt percussion**



Percussion Sounds

- Resonance: A hollow sound.
- Hyper resonance: A booming sound.
- Tympany: A musical sound or drum sound like that produced by the stomach.
- Dullness: Thud sound produced by dense structures such as the liver, and enlarged spleen, or a full bladder.
- Flatness: An extremely dull sound like that produced by very dense structures such as muscle or bone.

Percussion sounds

Sound	Intensity	Pitch	Length	Quality	Example of origin
Resonance (heard over part air and part solid	Loud	Low	Long	Hollow	Normal lung
Hyper-resonance (heard over mostly air	Very loud	Low	Long	Booming	Lung with emphysema
Tympany (heard over air)	Loud	High	Moderate	Drum like	Puffed-out cheek, gastric bubble
Dullness (heard over more solid tissue	Medium	Medium	Moderate	Thud like	Diaphragm, pleural effusion
Flatness (heard over very dense tissue	Soft	High	short	Flat	Muscle, Bone, Thigh

Assessment techniques
Auscultation

- Listening to sounds produced by the body

- Instrument: stethoscope (to skin)
  - Diaphragm –high pitched sounds
    - Heart
    - Lungs
    - Abdomen
  - Bell – low pitched sounds
    - Blood vessels





Assessment techniques - Setting

- Environment & Equipment



Technique

- General survey
- Head to toe or systems approach
- Minimize exposure
- Areas to assess first – unaffected areas, external before internal parts

Physical Health Exam-*General Survey*

- ***Appearance***
  - *Age, skin color, facial features*
  - *Body Structure - Stature, nutrition, posture, position, symmetry*
  - *Mobility - Gait, ROM*
- ***Behavior***
  - *Facial expression, mood/affect, speech, dress, hygiene*
- ***Cognition***
  - *Level of Consciousness and Orientation (x4)*
- *Include any signs of distress- facial grimacing, breathing problems*

Complete Health History



- **Biographical data**
- **Reason for Seeking Care**
- **History of Present Illness**
- Past Health
- Accidents and Injuries
- Hospitalizations and Operations
- Family History
- **Review of Systems**
- Functional Assessment (Activities of Daily Living)
- Perception of Health

Sources of Data

Data can be obtained from primary or secondary sources.

The primary source *of data* is the patient. In most instances the patient is considered to be the most accurate reporter. The alert and oriented patient can provide information about past illness and surgeries and present signs, symptoms, and lifestyle.

When the patient is unable to supply information because of deterioration of mental status, age, or seriousness of illness, secondary sources are used.

- *The Secondary sources of data* include family members, significant others, medical records, diagnostic procedures, ….
- Members of the patient's support system may be able to furnish information about the patient's past health status, current illness, allergies, and current medications.
- Other health team professionals are also helpful secondary sources (HCP)

Validating Data

The information gathered during the assessment phase must be complete, factual, and accurate because the nursing diagnosis and interventions are based on this information.

Assessment reduce potential risks associated with the administration of anesthesia

Validation is the act of "double-checking" or verifying data to confirm that it is accurate and factual.

Purposes of Data Validation

- Ensure that data collection is complete
- Ensure that objective and subjective data agree
- Obtain additional data that may have been overlooked
    - avoid jumping to conclusion
        - differentiate cues and inferences



Data Requiring Validation

Not every piece of data you collect must be verified. For example: you would not need to verify or repeat the client's pulse, temperature, or blood pressure unless certain conditions exist. Conditions that require data to be rechecked and validated include:

- Discrepancies or gaps between the subjective and objective data. For example, a male client tells you that he is very happy despite learning that he has terminal cancer.

Data Requiring Validation

- Discrepancies or gaps between what the client says at one time and then another time. For example, your female patient says she has never had surgery, but later in the interview she mentions that her appendix was removed at a military hospital when she was in the navy
    - Findings those are very abnormal and inconsistent with other findings. For example, the client has a temperature of 104°F degree. The client is resting comfortably. The client's skin is warm to touch and not flushed.

Methods of validation

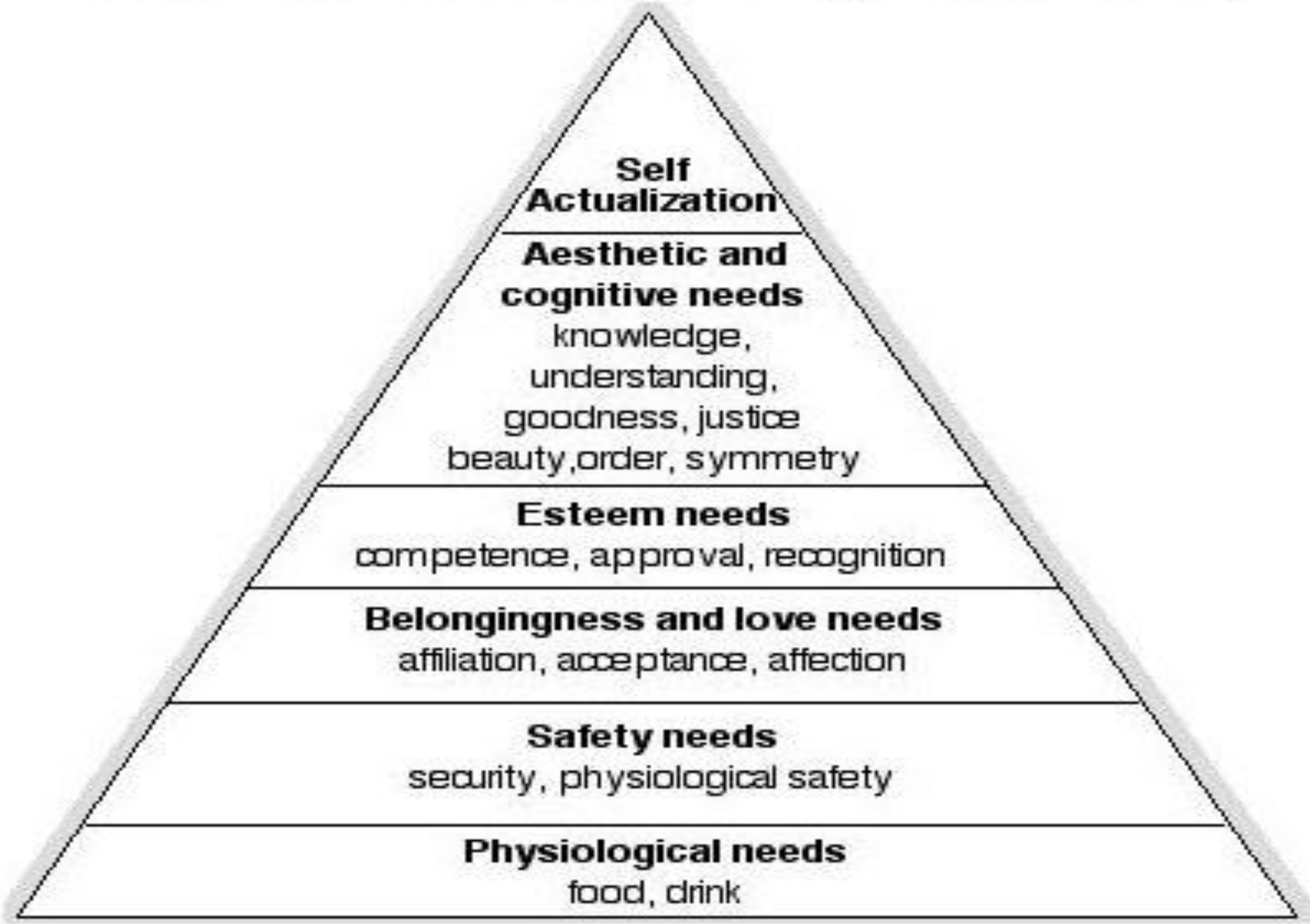
- Recheck your own data through a repeat assessment. For example, take the client's temperature again with a different thermometer.
- Clarify data with the client by asking additional questions. For example: if a client is holding his abdomen the nurse may assume he is having abdominal pain, when actually the client is very upset about his diagnosis and is feeling

- Verify the data with another health care professional. For example, ask a more experienced nurse to listen to the abnormal heart sounds you think you have just heard.
- Compare you objective findings with your subjective findings to uncover discrepancies. For example, if the client state that she “never gets any time in the sun” yet has dark, wrinkled, suntanned skin, you need to validate the client’s perception of never getting any time in the sun

Organizing data

We uses a written or computerized format that organizes the assessment data systematically. The format may be modified according to the client's physical status.

Maslow's Hierarchy of Needs



Body System Model

The Body systems model (also called the medical model or review of systems) focuses on the client's major anatomic systems. The framework allows nurses to collect data about past and present condition of each organ or body system and to examine thoroughly all body systems for actual and potential problems.

Gordon's Functional Health Patterns:

The client's strengths, talents and functional health patterns are an integral part of the assessment data. An assessment of functional health focuses on client's normal function and his or her altered function or risk for altered function.

- Health perception-health management pattern.
- Nutritional-metabolic pattern
- Elimination pattern

- Activity-exercise pattern
- Sleep-rest pattern
- Cognitive-perceptual pattern
- Self-perception-concept pattern
- Role-relationship pattern
- Sexuality-reproductive pattern
- Coping-stress tolerance pattern
- Value-belief pattern

Documenting Data:

To complete the assessment phase, the nurse records client's data.

Accurate documentation is essential and should include all data collected about the client's health status. Data are recorded in a factual manner and not interpreted by the nurse.

E.g.: the nurse record the client's breakfast intake as" coffee 240 mL. Juice 120 mL, 1 egg". Rather than as "appetite good".

Purposes of documentation

- Provides a chronological source of client assessment data and a progressive record of assessment findings that outline the client's course of care.
- Ensures that information about the client and family is easily accessible to members of the health care team; provides a vehicle for communication; and prevents fragmentation, repetition, and delays in carrying out the plan of care.
- Establishes a basis for screening or validation proposed diagnoses.
- Acts as a source of information to help diagnose new problems.

Purposes of documentation cont...

Offers a basis for determining the educational needs of the client, family, and significant others.

- Provides a basis for determining eligibility for care and reimbursement. Careful recording of data can support financial reimbursement or gain additional reimbursement for transitional or skilled care needed by the client.
- Constitutes a permanent legal record of the care that was or was not given to the client.
- Provides access to significant epidemiologic data for future investigations and research and educational endeavors.

Guidelines for documentation

- Document legibly or print neatly in unerasable ink
- Use correct grammar and spelling
- Avoid wordiness that creates redundancy
- Use phrases instead of sentences to record data
- Record data findings, not how they were obtained
- Write entries objectively without making premature judgments or diagnosis

Guidelines for documentation

- Record the client's understanding and perception of problems
- Avoid recording the word "normal" for normal findings
- Record complete information and details for all client symptoms or experiences
- Include additional assessment content when applicable
- Support objective data with specific observations obtained during the physical examination

VITAL SIGNS CHART استمارة العلامات الحيوية

| | | |

Assessment

- **Assessment is the first stage of the any health process in which the nurse should carry out a complete and holistic assessment.**
- **The purpose of this stage is to identify the patient's problems. These problems are expressed as either actual or potential. For example, a patient who has been rendered immobile by a road traffic accident may be assessed as having the "potential for impaired skin integrity related to immobility".**



Components of a nursing assessment

- **Biographic data – name, address, age, sex, martial status, occupation, religion.**
- **Reason for visit/Chief complaint – primary reason why client seek consultation or hospitalization.**
- **History of present Illness – includes: usual health status, chronological story, family history, disability assessment.**
- **Past Health History – includes all previous immunizations, experiences with illness.**
- **Family History – reveals risk factors for certain disease diseases (Diabetes, hypertension, cancer, mental illness).**

Components of a nursing assessment

- **Review of systems – review of all health problems by body systems**
- **Lifestyle – include personal habits, diets, sleep or rest patterns, activities of daily living, recreation or hobbies.**
- **Social data – include family relationships, ethnic and educational background, economic status, home and neighborhood conditions.**
- **Psychological data – information about the client's emotional state.**
- **Pattern of health care – includes all health care resources: hospitals, clinics, health centers, family doctors.**

Psychological And Social Examination

- Client's perception (why they think they have been referred/are being assessed; what they hope to gain from the meeting)
- Emotional health (mental health state, coping styles etc)
- Social health (accommodation, finances, relationships, genogram, employment status, ethnic back ground, support networks etc)
- Physical health (general health, illnesses, previous history, appetite, weight, sleep pattern, diurinal variations, alcohol, tobacco, street drugs; list any prescribed medication with comments on effectiveness)

Psychological And Social Examination

- Spiritual health (is religion important? If so, in what way? What/who provides a sense of purpose?)
- Intellectual health (cognitive functioning, hallucinations, delusions, concentration, interests, hobbies etc



Physical Examination

- A nursing assessment includes a physical examination: the observation or measurement of signs, which can be observed or measured, or symptoms such as nausea or vertigo, which can be felt by the patient.
- The techniques used may include Inspection, Palpation, Auscultation and Percussion in addition to the "vital signs" of temperature, blood pressure, pulse and respiratory rate, and further examination of the body systems such as the cardiovascular or musculoskeletal systems.

Assessment Tools

The index of independence in **activities of daily living**

- **Activities of daily living** (**ADLs**) are "the things we normally do in daily living including any daily activity we perform for self-care (such as feeding ourselves, bathing, dressing, grooming), work, homemaking, and leisure."

The Barthel Index

The Barthel Index consists of 10 items that measure a person's daily functioning specifically the activities of daily living and mobility. The items include feeding, moving from wheelchair to bed and return, grooming, transferring to and from a toilet, bathing, walking on level surface, going up and down stairs, dressing, continence of bowels and bladder.



Patient Name: _____ **Rater:** _____ **Date:** / / :

Activity	Score
Feeding 0 = unable 5 = needs help cutting, spreading butter, etc., or requires modified diet 10 = independent	0 5 10
Bathing 0 = dependent 5 = independent (or in shower)	0 5
Grooming 0 = needs to help with personal care 5 = independent face/hair/teeth/shaving (implements provided)	0 5
Dressing 0 = dependent 5 = needs help but can do about half unaided 10 = independent (including buttons, zips, laces, etc.)	0 5 10
Bowels 0 = incontinent (or needs to be given enemas) 5 = occasional accident 10 = continent	0 5 10

Bladder 0 = incontinent, or catheterized and unable to manage alone 5 = occasional accident 10 = continent	0 5 10
Toilet Use 0 = dependent 5 = needs some help, but can do something alone 10 = independent (on and off, dressing, wiping)	0 5 10
Transfers (bed to chair and back) 0 = unable, no sitting balance 5 = major help (one or two people, physical), can sit 10 = minor help (verbal or physical) 15 = independent	0 5 10 15
Mobility (on level surfaces) 0 = immobile or < 50 yards 5 = wheelchair independent, including corners, > 50 yards 10 = walks with help of one person (verbal or physical) > 50 yards 15 = independent (but may use any aid; for example, stick) > 50 yards	0 5 10 15
Stairs 0 = unable 5 = needs help (verbal, physical, carrying aid) 10 = independent	0 5 10
TOTAL (0 - 100)	_____

Cont…..

- The general health **questionnaire**
- **Mental health status** examination

The Mental Status Exam (MSE) is a series of questions and observations that provide a snapshot of a client's current mental, cognitive, and behavioral condition.

Glasgow Coma Scale

Best eye response (E)	Spontaneous--open with blinking at baseline	4
	Opens to verbal command, speech, or shout	3
	Opens to pain, not applied to face	2
	None	1
Best verbal response (V)	Oriented	5
	Confused conversation, but able to answer questions	4
	Inappropriate responses, words discernible	3
	Incomprehensible speech	2
	None	1
Best motor response (M)	Obeys commands for movement	6
	Purposeful movement to painful stimulus	5
	Withdraws from pain	4
	Abnormal (spastic) flexion, decorticate posture	3
	Extensor (rigid) response, decerebrate posture	2
	None	1

Conclusion

Assessment is the first and most critical step of nursing process. Accuracy of assessment data affects all other phases of the nursing process. A complete data base of both subjective and objective data allows the nurse to formulate nursing diagnosis, develop client goals, and intervenes to promote heath and prevent disease.



The Skin, Hair and Nails
Assessment

Common or Concerning Symptoms

- Hair loss
- Rash
- Moles

HEALTH PROMOTION AND COUNSELING

Important Topics for Health Promotion and Counseling

- Risk factors for melanoma
- Avoidance of excessive sun exposure

Counsel patient to avoid unnecessary sun exposure and to use sunscreen with at least SPF-15. Teach the ABCD screen for dysplastic nevi/melanomas: **A**symmetry, irregular **B**orders, variation in **C**olor, and **D**iameter >6 mm. Survey skin at 3-year intervals for patients 20 to 40 years of age and annually for patients older than 40 years.

TECHNIQUES OF EXAMINATION

EXAMINATION TECHNIQUES	POSSIBLE FINDINGS
SKIN	
Examine the entire skin surface under good lighting. **Inspect** and **palpate** each area.	
Note:	
• Color	Cyanosis, jaundice, carotenemia, changes in melanin
• Moisture	Dry, oily
• Temperature	Cool, warm
• Texture	Smooth, rough

• Mobility—ease with which a fold of skin can be moved	Decreased in edema
• Turgor—speed with which the fold returns into place	Decreased in dehydration
Note any lesions and their	
• Anatomical location and distribution	Generalized, localized
• Patterns and shapes	Linear, clustered, dermatomal
• Type	Macule, papule, pustule, bulla, tumor
• Color	Red, white, brown, heliotrope

EXAMINATION TECHNIQUES	POSSIBLE FINDINGS

HAIR

Inspect and **palpate** the hair.

Note

• Quantity	Thin, thick
• Distribution	Patchy or total alopecia
• Texture	Fine, coarse

NAILS

Inspect and **palpate** the fingernails and toenails.

Note

- Color — Cyanosis, pallor
- Shape — Clubbing
- Any lesions — Paronychia, onycholysis





TABLE 6-2 Primary Skin Lesions

Flat, Nonpalpable Lesions With Changes in Skin Color



Macule—Small flat spot, up to 1.0 cm



Examples: Hemangioma, vitiligo



Patch—Flat spot, 1.0 cm or larger
Example: Café-au-lait spot

Palpable Elevations: Solid Masses



Papule—Up to 1.0 cm
Example: An elevated nevus



Plaque—Elevated superficial lesion 1.0 cm or larger, often formed by coalescence of papules

TABLE 6-2 Primary Skin Lesions (continued)



Example: Psoriasis



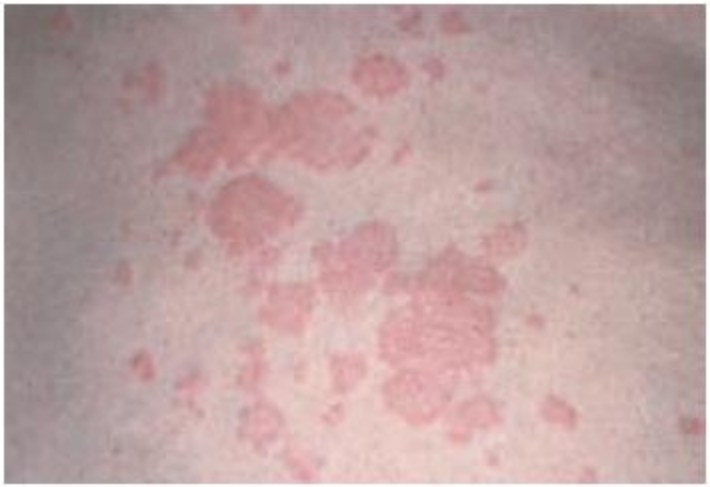
Nodule—Marble-like lesion larger than 0.5 cm, often deeper and firmer than a papule

Example: Dermatofibroma



Cyst—Nodule filled with expressible material, either liquid or semisolid

Example: Epidermal inclusion cyst



Wheal—A somewhat irregular, relatively transient, superficial area of localized skin edema

Examples: Mosquito bite, hives (urticaria)

TABLE 6-2 Primary Skin Lesions (continued)

Palpable Elevations With Fluid-Filled Cavities



Vesicle—Up to 1.0 cm; filled with serous fluid



Example: Herpes simplex



Bulla—1.0 cm or larger; filled with serous fluid



Example: Insect bite



Pustule—Filled with pus

TABLE 6-2 Primary Skin Lesions (continued)



Examples: Acne, small pox (impetigo)



Burrow (scabies)—A minute, slightly raised tunnel in the epidermis, commonly found on the finger webs and on the sides of the fingers. It looks like a short (5–15 mm), linear or curved gray line and may end in a tiny vesicle. With a magnifying lens, look for the *burrow* of the mite that causes scabies.

TABLE 6-3 Secondary Skin Lesions

May arise from primary lesions, overtreatment, excess scratching



Scale—A thin flake of dead, exfoliated epidermis



Examples: Ichthyosis, dandruff, dry skin, psoriasis



Scars—Connective tissue that arises from injury or disease
Example: Acne



Keloids—Hypertrophic scarring that extends beyond the borders of the initiating injury



Fissure—A linear crack in the skin, often resulting from excessive dryness
Example: Athlete's foot



Ulcer—A deeper loss of epidermis and dermis; may bleed and scar
Examples: Stasis ulcer of venous insufficiency, syphilitic chancre



Kaposi's Sarcoma in AIDS

May appear in many forms: macules, papules, plaques, or nodules almost anywhere on the body. Lesions are often multiple and may involve internal structures.

TABLE 6-7 Characteristics of Nevi (Moles)

Normal



Diameter smaller than 6 mm
Symmetric; regular borders; even in color

Malignant Melanoma: "ABCD"



Asymmetric



Borders irregular



Color varied
Diameter more than 6 mm

TABLE 6-8 Hair Loss





Alopecia Areata

Clearly demarcated round or oval patches of hair loss, usually affecting young adults and children. There is no visible scaling or inflammation.

TABLE 6-9 Fingernails

Clubbing



Dorsal phalanx rounded and bulbous; convexity of nail plate increased. Angle between plate and proximal nail fold increased to 180° or more. Proximal nail folds feel spongy. Many causes, including chronic hypoxia and lung cancer.

Paronychia



Inflammation of proximal and lateral nail folds, acute or chronic. Folds red, swollen, may be tender.



Thorax and Lung Assessment

Chapter (8)

THE HEALTH HISTORY

Common or Concerning Symptoms

- Chest pain
- Shortness of breath (dyspnea)
- Wheezing
- Cough
- Blood-streaked sputum (hemoptysis)

Complaints of *chest pain* or *chest discomfort* raise the specter of heart disease but often arise from conditions in the thorax and lungs. For this important symptom, keep the possible causes below in mind. Also see Table 8-1, Chest Pain, pp. 152–153.

- The myocardium — Angina pectoris, myocardial infarction
- The pericardium — Pericarditis
- The aorta — Dissecting aortic aneurysm

• The trachea and large bronchi	Bronchitis
• The parietal pleura	Pericarditis, pneumonia
• The chest wall, including the musculoskeletal system and skin	Costochondritis, herpes zoster

- The esophagus — Reflux esophagitis, esophageal spasm
- Extrathoracic structures such as the neck, gallbladder, stomach — Cervical arthritis, biliary colic, gastritis

For patients who are *short of breath*, focus on such *pulmonary complaints* as

- dyspnea and wheezing — See Table 8-2, Dyspnea, pp. 154–155.
- cough and hemoptysis — See Table 8-3, Cough and Hemoptysis, pp. 156–159.

HEALTH PROMOTION AND COUNSELING

Despite declines in smoking over the past several decades, ~23% of Americans still smoke.* Regularly counsel all adults, pregnant women, parents, and adolescents who smoke to stop. Include "the five **A's**":

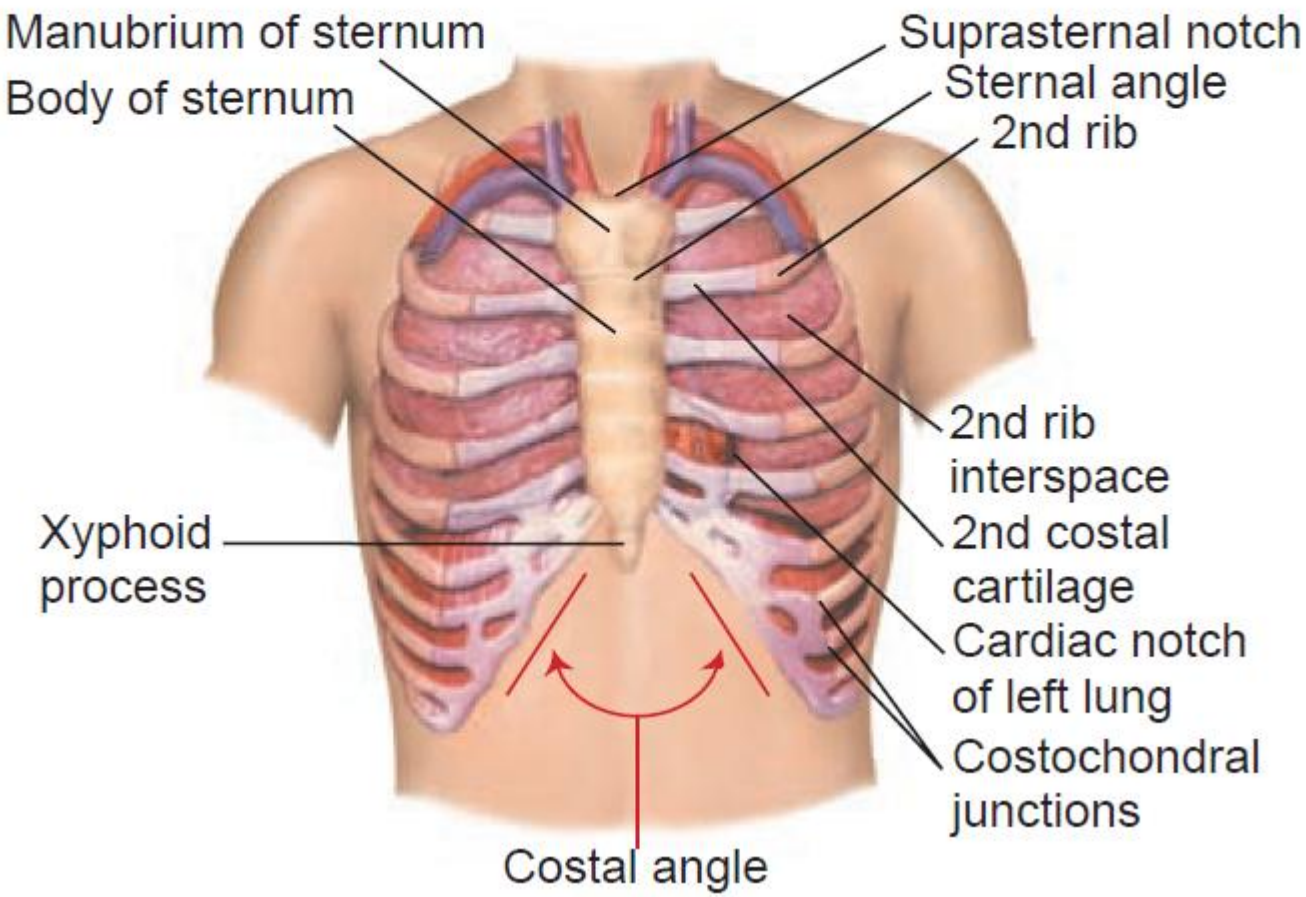
- **A**sk about smoking at each visit,
- **A**dvise patients to quit in regular, clear, personalized messages.
- **A**ssess patient readiness to quit.
- **A**ssist patients to set stop dates and provide educational materials for self-help.
- **A**rrange for follow-up visits to monitor and support progress.

Provide flu shots and pneumonia vaccine in target groups.

TECHNIQUES OF EXAMINATION

EXAMINATION TECHNIQUES | POSSIBLE FINDINGS

SURVEY OF THORAX



Inspect the thorax and its respiratory movements.
Note:

• Rate, rhythm, depth, and effort of breathing	Tachypnea, hyperpnea, Cheyne–Stokes breathing
• Inspiratory retraction of the supraclavicular areas	Occurs in chronic obstructive pulmonary disease (COPD), asthma, upper airway obstruction
• Inspiratory contraction of the sternomastoids	Indicates severe breathing difficulty

Observe shape of patient's chest.	Normal or barrel chest (see Table 8-4, Deformities of the Thorax, pp. 159–160)
Listen to patient's breathing for	
• Rate and rhythm of breathing	14–16 breaths/minute in adults (see Chapter 4, pp. 64–65)

- Stridor

  Stridor in upper airway obstruction from foreign body or epiglottitis

- Wheezes

  Expiratory wheezing in asthma and COPD

THE POSTERIOR CHEST

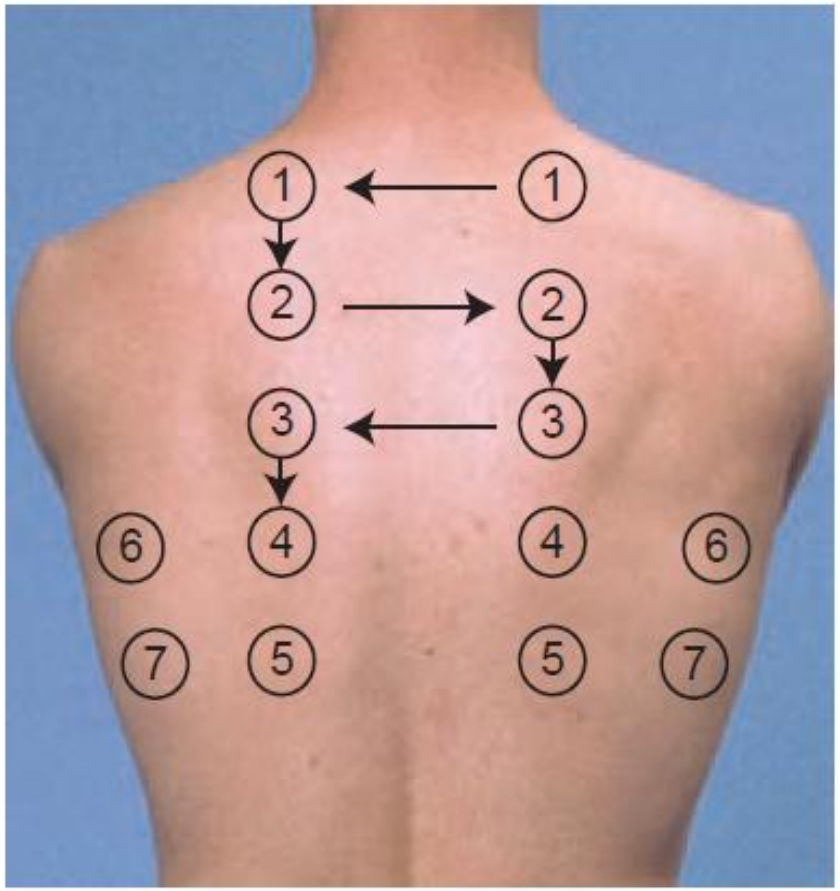
Inspect the chest for

• Deformities or asymmetry	Kyphoscoliosis
• Abnormal inspiratory retraction of the interspaces	Retraction in airway obstruction
• Impairment or unilateral lag in respiratory movement	Disease of the underlying lung or pleura, phrenic nerve palsy

- Tender areas — Fractured ribs
- Assessment of visible abnormalities — Masses, sinus tracts
- Chest expansion — Impairment, both sides in COPD and restrictive lung disease



EXAMINATION TECHNIQUES	POSSIBLE FINDINGS
• Tactile fremitus as the patient says "aa" or "blue moon"	Local or generalized decrease or increase
Percuss the chest in the areas illustrated, comparing one side with the other at each level, using the side-to-side "ladder pattern."	*Dullness* when fluid or solid tissue replaces normally air-filled lung; *hyperresonance* in emphysema or pneumothorax


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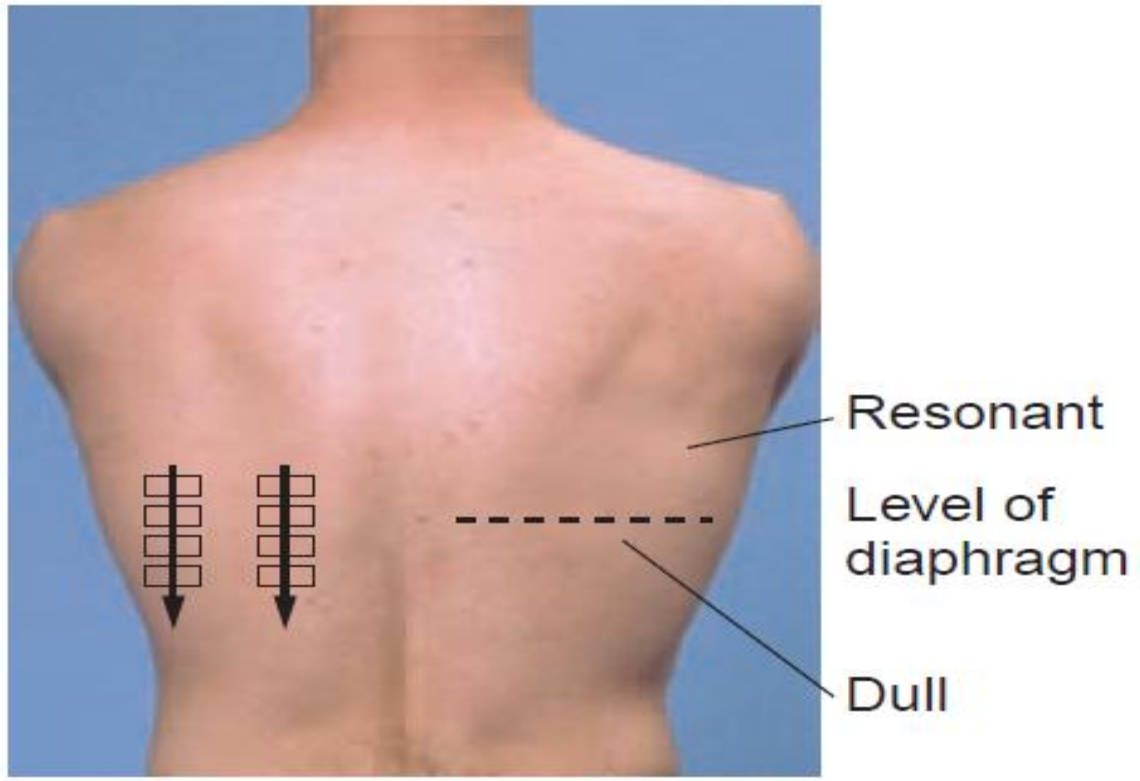


Percussion Notes

	Relative Intensity, Pitch, and Duration	Examples
Flatness	Soft/high/short	Large pleural effusion
Dullness	Medium/medium/medium	Lobar pneumonia
Resonance	Loud/low/long	Normal lung, simple chronic bronchitis
Hyperresonance	Louder/lower/longer	Emphysema, pneumothorax
Tympany	Loud/high (timbre is musical)	Large pneumothorax

Percuss level of diaphragmatic dullness on each side and **estimate** diaphragmatic descent after patient takes full inspiration.

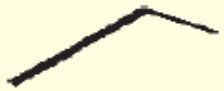
Pleural effusion or a *paralyzed diaphragm* raises level of dullness.



Listen to chest with stethoscope in the "ladder" pattern, again comparing sides.	See Table 8-5, Physical Findings in Selected Chest Disorders, p. 161.
• Evaluate the breath sounds.	Vesicular, bronchovesicular, or bronchial breath sounds; decreased breath sounds from decreased airflow
• Note any adventitious (added) sounds.	Crackles (fine and coarse) and continuous sounds (wheezes and rhonchi)

Observe their qualities, place in the respiratory cycle, and location on the chest wall. Do they clear with deep breathing or coughing?

Breath Sounds

	Duration	Intensity and Pitch of Expiratory Sound	Example Locations
Vesicular 	Insp > exp	Soft/low	Most of the lungs
Bronchovesicular 	Insp = exp	Medium/ medium	1st and 2nd interspaces, interscapular area
Bronchial 	Exp > insp	Loud/high	Over the manubrium; lobar pneumonia
Tracheal 	Insp = exp	Very loud/ high	Over the trachea

Adventitious or Added Breath Sounds

Crackles (or Rales)	Wheezes and Rhonchi
• **Discontinuous**	• **Continuous**
• Intermittent, nonmusical, and brief	• ≥250 msec, musical, prolonged (but not necessarily persisting throughout the respiratory cycle)
	• Like dashes in time
• Like dots in time	• *Wheezes:* Relatively high-pitched (≥400 Hz) with hissing or shrill quality
• *Fine crackles:* Soft, high-pitched, very brief (5–10 msec)	
pulmonary oedema or interstitial fibrosis	
• *Coarse crackles:* Somewhat louder, lower in pitch, brief (20–30 msec)	• *Rhonchi:* Relatively low-pitched (≤200 Hz) with snoring quality
bronchiectasis or resolving pneumonia	

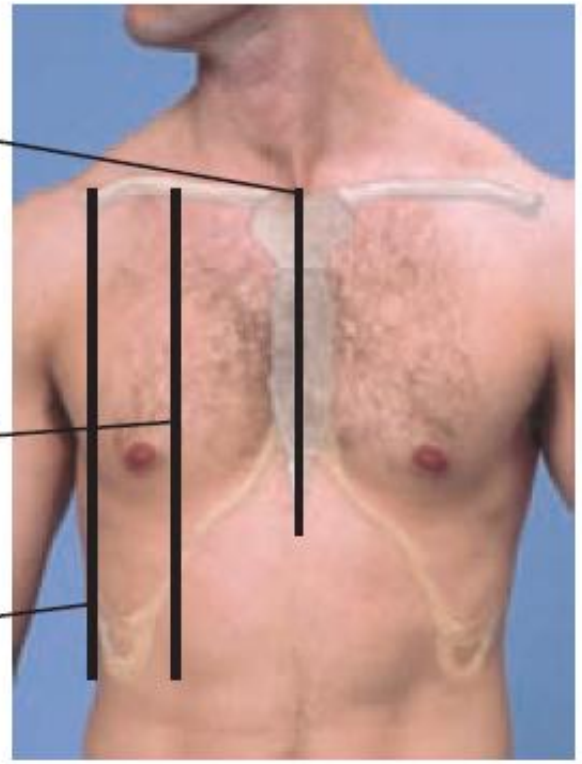
EXAMINATION TECHNIQUES	POSSIBLE FINDINGS
Assess transmitted voice sounds if you have heard bronchial breath sounds in abnormal places. Ask patient to	
• Say “99” and “ee.”	Bronchophony and egophony
• Whisper “99” or “1, 2, 3.”	Whispered pectoriloquy

Transmitted Voice Sounds

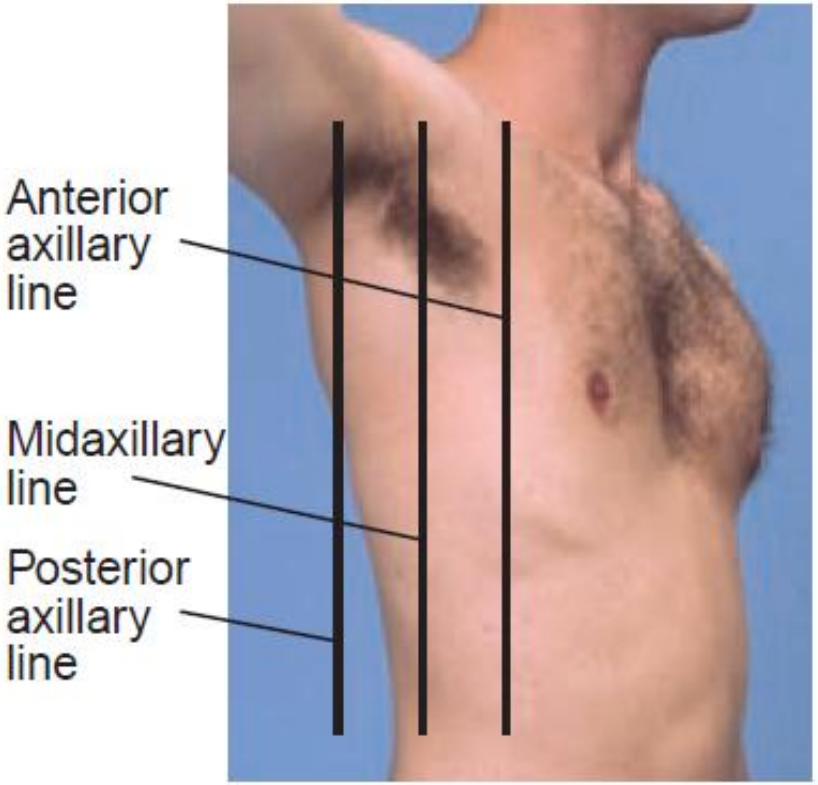
Through Normally Air-Filled Lung	Through Airless Lung*
Spoken words muffled and indistinct	Spoken words louder, clearer (*bronchophony*)
Spoken "ee" heard as "ee"	Spoken "ee" heard as "ay" (*egophony*)
Whispered words faint and indistinct, if heard at all	Whispered words louder, clearer (*whispered pectoriloquy*)
Usually accompanied by vesicular breath sounds and normal tactile fremitus	Usually accompanied by bronchial or bronchovesicular breath sounds and increased tactile fremitus

*As in lobar pneumonia and toward the top of a large pleural effusion

THE ANTERIOR CHEST



ANTERIOR VIEW



RIGHT ANTERIOR OBLIQUE VIEW

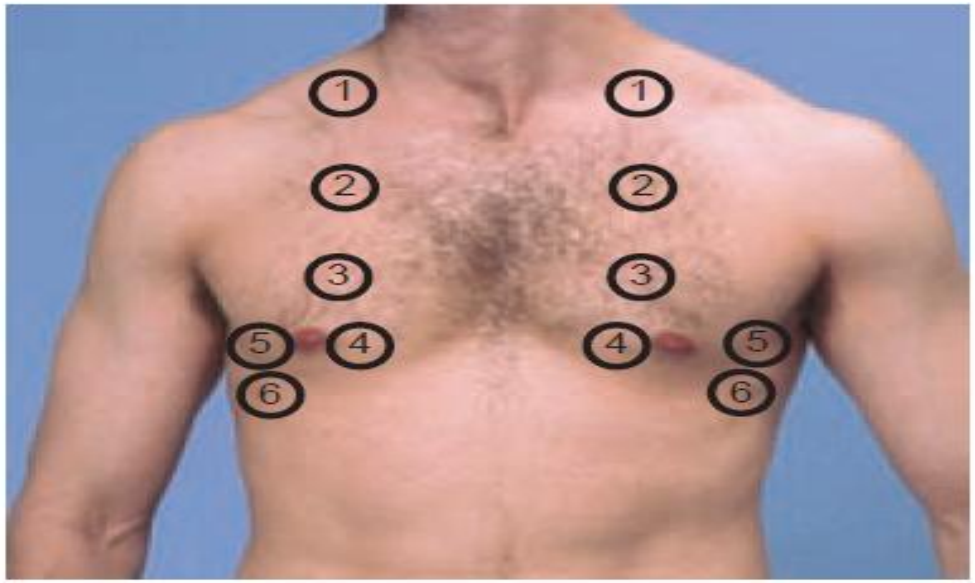
Inspect the chest for

• Deformities or asymmetry	Pectus excavatum
• Intercostal retraction	From obstructed airways
• Impaired or lagging respiratory movement	Disease of the underlying lung or pleura, phrenic nerve palsy

Palpate the chest for

• Tender areas	Tender pectoral muscles, costochondritis
• Assessment of visible abnormalities	Flail chest
• Respiratory expansion	

EXAMINATION TECHNIQUES



Listen to the chest with stethoscope. **Note:**

- Breath sounds
- Adventitious sounds
- If indicated, transmitted voice sounds

POSSIBLE FINDINGS

ASSESSMENT OF PULMONARY FUNCTION

Walk with patient down the hall or up a flight of stairs. Observe rate, effort, and sound of breathing, and inquire about symptoms.

Older adults walking 8 feet in <3 seconds are less likely to be disabled than those taking >5–6 seconds.

FORCED EXPIRATORY TIME

Ask patient to take a deep breath in and then breathe out as quickly and completely as possible, with mouth open. Listen over trachea with diaphragm of stethoscope, and time audible expiration. Try to get three consistent readings, allowing rests as needed.

If the patient understands and cooperates well, a forced expiratory time of 6 to 8 strongly suggests COPD.

RECORDING YOUR FINDINGS

Recording the Physical Examination— The Thorax and Lungs

"Thorax is symmetric with good expansion. Lungs resonant. Breath sounds vesicular; no rales, wheezes, or rhonchi. Diaphragms descend 4 cm bilaterally."

OR

"Thorax symmetric with moderate kyphosis and increased anteroposterior (AP) diameter, decreased expansion. Lungs are hyperresonant. Breath sounds distant with delayed expiratory phase and scattered expiratory wheezes. Fremitus decreased; no bronchophony, egophony, or whispered pectoriloquy. Diaphragms descend 2 cm bilaterally."

(Suggests COPD)

TABLE 8-1 Chest Pain

Problem and Location	Quality, Severity, Timing, and Associated Symptoms
Cardiovascular	
Angina Pectoris Retrosternal or across the anterior chest, sometimes radiating to the shoulders, arms, neck, lower jaw, or upper abdomen	**Quality:** Pressing, squeezing, tight, heavy, occasionally burning **Severity:** Mild to moderate, sometimes perceived as discomfort rather than pain **Timing:** Usually 1–3 min but up to 10 min; prolonged episodes up to 20 min **Associated Symptoms:** Sometimes dyspnea, nausea, swelling
Myocardial Infarction Same as in angina	**Quality:** Same as in angina **Severity:** Often but not always a severe pain **Timing:** 20 min to several hours **Associated Symptoms:** Nausea, vomiting, sweating, weakness

Pericarditis *Precordial:* May radiate to the tip of the shoulder and to the neck	**Quality:** Sharp, knifelike **Severity:** Often severe **Timing:** Persistent **Associated Symptoms:** Of the underlying illness; relieved by leaning forward
Retrosternal	**Quality:** Crushing **Severity:** Severe **Timing:** Persistent **Associated Symptoms:** Of the underlying illness

Dissecting Aortic Aneurysm

Anterior chest, radiating to the neck, back, or abdomen

Quality: Ripping, tearing
Severity: Very severe
Timing: Abrupt onset, early peak, persistent for hours or more
Associated Symptoms: Syncope, hemiplegia, paraplegia

TABLE 8-1 Chest Pain (continued)

Problem and Location	Quality, Severity, Timing, and Associated Symptoms
Pulmonary	
Tracheobronchitis Upper sternal or on either side of the sternum	**Quality:** Burning **Severity:** Mild to moderate **Timing:** Variable **Associated Symptoms:** Cough
Pleural Pain Chest wall overlying the process	**Quality:** Sharp, knifelike **Severity:** Often severe **Timing:** Persistent **Associated Symptoms:** Of the underlying illness

Gastrointestinal and Other

Reflex Esophagitis Retrosternal, may radiate to the back	**Quality:** Burning, may be squeezing **Severity:** Mild to severe **Timing:** Variable **Associated Symptoms:** Sometimes regurgitation, dysphagia
Diffuse Esophageal Spasm Retrosternal, may radiate to the back, arms, and jaw	**Quality:** Usually squeezing **Severity:** Mild to severe **Timing:** Variable **Associated Symptoms:** Dysphagia

Chest Wall Pain

Often below the left breast or along the costal cartilages; also elsewhere

Quality: Stabbing, sticking, or dull aching
Severity: Variable
Timing: Fleeting to hours or days
Associated Symptoms: Often local tenderness

Anxiety/Panic Attack

TABLE 8-2 Dyspnea

Problem	Timing	Provoking and Relieving Factors
Left-sided Heart Failure *(left ventricular failure or mitral stenosis)*	Dyspnea may progress slowly or suddenly, as in acute pulmonary edema	↑ by exertion, lying down ↓ by rest, sitting up, though dyspnea may become persistent **Associated Symptoms:** Often cough, orthopnea, paroxysmal nocturnal dyspnea, sometimes wheezing

Chronic Bronchitis *(may be seen with COPD)*	Chronic productive cough followed by slowly progressive dyspnea	↑ by exertion, inhaled irritants, respiratory infections ↓ by expectoration, rest though dyspnea may become persistent **Associated Symptoms:** Chronic productive cough, recurrent respiratory infections; wheezing possible

Chronic Obstructive Pulmonary Disease (COPD)	Slowly progressive; relatively mild cough later	↑ by exertion ↓ by rest, though dyspnea may become persistent **Associated Symptoms:** Cough with scant mucoid sputum
Asthma	Acute episodes, then symptom-free periods; nocturnal episodes common	↑ by allergens, irritants, respiratory infections, exercise, emotion ↓ by separation from aggravating factors **Associated Symptoms:** Wheezing, cough, tightness in chest

The Abdomen Assessment

Chapter (11)

The Health History

Common or Concerning Symptoms

Gastrointestinal Disorders

- Abdominal pain, acute and chronic
- Indigestion, nausea, vomiting including blood, loss of appetite, early satiety
- Dysphagia and/or odynophagia
- Change in bowel function
- Diarrhea, constipation
- Jaundice

Urinary and Renal Disorders

- Suprapubic pain
- Dysuria, urgency, or frequency
- Hesitancy, decreased stream in males
- Polyuria or nocturia
- Urinary incontinence
- Hematuria
- Kidney or flank pain
- Ureteral colic

PATTERNS AND MECHANISMS OF ABDOMINAL PAIN

Be familiar with three broad categories:

Visceral pain—occurs when hollow abdominal organs such as the intestine or biliary tree contract unusually forcefully or are distended or stretched.

Visceral pain in the right upper quadrant (RUQ) from liver distention against its capsule in *alcoholic hepatitis*

- May be difficult to localize
- Varies in quality; may be gnawing, burning, cramping, or aching
- When severe, may be associated with sweating, pallor, nausea, vomiting, restlessness.

Parietal pain—from inflammation of the parietal peritoneum

- Steady, aching
- Usually more severe
- Usually more precisely localized over the involved structure than visceral pain

Visceral periumbilical pain in *early acute appendicitis* from distention of inflamed appendix gradually changes to parietal pain in the right lower quadrant (RLQ) from inflammation of the adjacent parietal peritoneum.

Referred pain—occurs in more distant sites innervated at approximately the same spinal levels as the disordered structure.

Pain of duodenal or pancreatic origin may be referred to the back; pain from the biliary tree—to the right shoulder or right posterior chest.

Pain from the chest, spine, or pelvis may be referred to the abdomen.

Pain from *pleurisy* or *acute myocardial infarction* may be referred to the upper abdomen.

THE GASTROINTESTINAL TRACT

Ask patients to *describe the abdominal pain in their own words*, especially timing of the pain (acute or chronic); then ask them to *point to the pain*.

Pursue important details: "Where does the pain start?"

“Does it radiate or travel?”
“What is the pain like?”
“How severe is it?” “How about on a scale of 1 to 10?”
“What makes it better or worse?”

Elicit any *symptoms associated with the pain*, such as fever or chills; ask their sequence.

Upper Abdominal Pain or Discomfort. Ask about chronic or recurrent upper abdominal discomfort, or *dyspepsia*. Related symptom include bloating, nausea, upper abdominal fullness, and heartburn.

Find out just what your patient means. Possibilities include:

- Bloating from excessive gas, especially with frequent belching, abdominal distention, or flatus, the passage of gas by rectum

- *Nausea and vomiting*

- Unpleasant *abdominal fullness* after normal meals or *early satiety*, the inability to eat a full meal

  Consider diabetic gastroparesis, anticholinergic drugs, gastric outlet obstruction, gastric cancer. Early satiety may signify *hepatitis.*

- *Heartburn*

  Suggests *gastroesophageal reflux disease (GERD)*

Lower Abdominal Pain or Discomfort

If acute, is the pain sharp and continuous or intermittent and cramping?

Right lower quadrant (RLQ) pain, or pain migrating from periumbilical region in *appendicitis;* in women with RLQ pain, possible *pelvic inflammatory disease, ectopic pregnancy*

Left lower quadrant (LLQ) pain in *diverticulitis*

If chronic, is there a change in bowel habits? Alternating diarrhea and constipation?

Colon cancer; irritable bowel syndrome

Other GI Symptoms

- Anorexia

Liver disease, pregnancy, diabetic ketoacidosis, adrenal insufficiency, uremia, anorexia nervosa

- Dysphagia or difficulty swallowing

Of solids and liquids: usually neuromuscular disorders affecting motility

• Odynophagia, or painful swallowing	Radiation; caustic ingestion, infection from *cytomegalovirus,* herpes simplex, HIV
• Diarrhea, acute (<2 weeks) and chronic	Acute infection (viral, salmonella, shigella, etc.); chronic in *Crohn's disease,* ulcerative colitis; if oily (*steatorrhea*)—pancreatic insufficiency. See Table 11-1.

• Constipation	Medications, especially anticholinergic agents and opioids; colon cancer
• Melena, or black tarry stools	GI bleed
• Jaundice from increased levels of bilirubin; intrahepatic jaundice can be *hepatocellular*, from damage to the hepatocytes, or *cholestatic*, from impaired excretion caused by damaged hepatocytes or intra-hepatic bile ducts	Impaired excretion of conjugated bilirubin in *viral hepatitis, cirrhosis, primary biliary cirrhosis,* drug-induced cholestasis

Extrahepatic jaundice from obstructed extrahepatic bile ducts, commonly the cystic and common bile ducts

Ask about the *color of the urine and stool.*

Dark urine from increased conjugated bilirubin in serum excreted in urine; acholic clay-colored stool when excretion of bilirubin into intestine is obstructed

Risk Factors for Liver Disease

- *Hepatitis A:* Travel or meals in areas with poor sanitation, ingestion of contaminated water or foodstuffs
- *Hepatitis B:* Parenteral or mucous membrane exposure to infectious body fluids such as blood, serum, semen, and saliva, especially through sexual contact with an infected partner or use of shared needles for injection drug use
- *Hepatitis C:* Illicit intravenous drug use or blood transfusion
- *Alcoholic hepatitis* or *alcoholic cirrhosis:* Interview the patient carefully about alcohol use

(*continued*)

Risk Factors for Liver Disease (continued)

- *Toxic liver damage* from medications, industrial solvents, or environmental toxins
- *Extrahepatic biliary obstruction:* Resulting from gallbladder disease or surgery
- *Hereditary disorders* in the Family History

THE URINARY TRACT

Ask about *pain on urination*, usually a burning sensation, sometimes termed *dysuria* (also refers to difficulty voiding).

Bladder infection

Also, consider bladder stones, foreign bodies, tumors, and *acute prostatitis*. In women, internal burning in *urethritis*, external burning in *vulvovaginitis*

Other associated symptoms include

- *Urgency*, an unusually intense and immediate desire to void

May lead to urge incontinence

- *Urinary frequency*, or abnormally frequent voiding
- Fever or chills; blood in the urine
- Any pain in the abdomen, flank, or back

Dull, steady pain in *pyelonephritis;* severe colicky pain in ureteral obstruction from renal stone

In men *hesitancy* in starting the urine stream, *straining to void, reduced caliber and force of the urine stream*, or *dribbling* as they complete voiding

Prostatitis, urethritis

Assess any:

• *Polyuria*, a significant increase in 24-hour urine volume	*Diabetes mellitus, diabetes insipidus*
• *Nocturia*, urinary frequency at night	Bladder obstruction
• *Urinary incontinence*, involuntary loss of urine	See Table 11-2, Urinary Incontinence, pp. 211–212.
• From coughing, sneezing, lifting	*Stress incontinence* (poor urethral sphincter tone)
• From urge to void	*Urge incontinence* (detrusor overactivity)
• From bladder fullness with leaking but incomplete emptying	*Overflow incontinence* (anatomic obstruction, impaired neural innervation to bladder)

HEALTH PROMOTION AND COUNSELING

Important Topics for Health Promotion and Counseling

- Screening for alcohol abuse
- Risk factors for hepatitis A, B, and C
- Screening for colon cancer

Assessing *use of alcohol* is an important clinician responsibility. Focus on detection, counseling, and, for significant impairment, specific treatment recommendations. Use the four CAGE questions to screen for alcohol dependence or abuse in all adolescents and adults, including pregnant women (see Chap. 3, p. 41). Brief counseling interventions have been shown to reduce alcohol consumption by 13% to 34% over 6 to 12 months.

Protective measures against *infectious hepatitis* include counseling about transmission:

- *Hepatitis A:* Transmission is fecal–oral. Illness occurs approximately 30 days after exposure. Hepatitis A vaccine is recommended for travelers to endemic areas; food handlers; military personnel; caretakers of children; Native Americans and Alaska Natives; selected health care, sanitation, and laboratory workers; homosexual men; and injection drug users.

- *Hepatitis B:* Transmission occurs during contact with infected body fluids, such as blood, semen, saliva, and vaginal secretions. Infection increases risk of fulminant hepatitis, chronic infection, and subsequent cirrhosis and hepatocellular carcinoma. Provide counseling and serologic screening for patients at risk. Hepatitis B vaccine is recommended for all young adults not previously immunized, injection drug users and their sexual partners, people at risk for sexually transmitted diseases, travelers to endemic areas, recipients of blood products as in hemodialysis, and health care workers with frequent exposure to blood products. Many of these groups also should be screened for HIV infection, especially pregnant women at their first prenatal visit.

- *Hepatitis C:* Hepatitis C, now the most common form, is spread by blood exposure and is associated with injection drug use. No vaccine is available.

For colorectal cancer, screen for risk factors, namely family history of colonic polyps, history of colorectal cancer or adenoma in a first-degree relative, and personal history of ulcerative colitis, adenomatous polyps, or prior diagnosis of endometrial, ovarian, or breast cancer. For men and women at average risk, the American Cancer Society recommends *one* of the following beginning at age 50:

- Fecal occult blood test (FOBT) annually
- Flexible sigmoidoscopy every 5 years
- Annual FOBT plus flexible sigmoidoscopy every 5 years
- Double contrast barium enema every 5 years
- Colonoscopy every 10 years

TECHNIQUES OF EXAMINATION

EXAMINATION TECHNIQUES	POSSIBLE FINDINGS
THE ABDOMEN	
Inspect the abdomen, including	
• Skin	Scars, striae, veins
• Umbilicus	Hernia, inflammation
• Contours for shape, symmetry, enlarged organs or masses	Bulging flanks, suprapubic bulge, large liver or spleen, tumors
• Any peristaltic waves	GI obstruction
• Any pulsations	Increased in aortic aneurysm
Auscultate the abdomen for	
• Bowel sounds	Increased or decreased motility

- Bruits — Bruit of renal artery stenosis
- Friction rubs — Liver tumor, splenic infarct

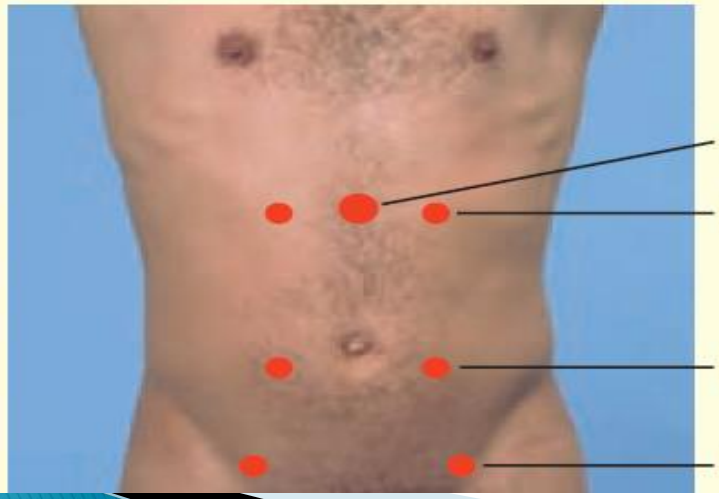
Bowel Sounds and Bruits

Change	Seen With
Increased bowel sounds	Diarrhea Early intestinal obstruction
Decreased, then absent bowel sounds	Adynamic ileus Peritonitis
High-pitched tinkling bowel sounds	Intestinal fluid Air under tension in a dilated bowel

(continued)

Bowel Sounds and Bruits (continued)

Change	Seen With
High-pitched rushing bowel sounds with cramping	Intestinal obstruction
Hepatic bruit	Carcinoma of the liver Alcoholic hepatitis
Arterial bruits	Partial obstruction of the aorta or renal arteries



Percuss the abdomen for patterns of tympany and dullness.

Ascites, GI obstruction, pregnant uterus, ovarian tumor

Palpate all quadrants of the abdomen

See Table 11-3.

- Lightly for guarding, rebound, and tenderness



Firm, boardlike abdominal wall suggests peritoneal inflammation. *Guarding* occurs when the patient flinches, grimaces, or reports pain during palpation.

Rebound tenderness from peritoneal inflammation; pain is greater when you withdraw

EXAMINATION TECHNIQUES

POSSIBLE FINDINGS

your hand than when you press down. Press slowly on a tender area, then quickly "let go."

- Deeply for masses or tenderness

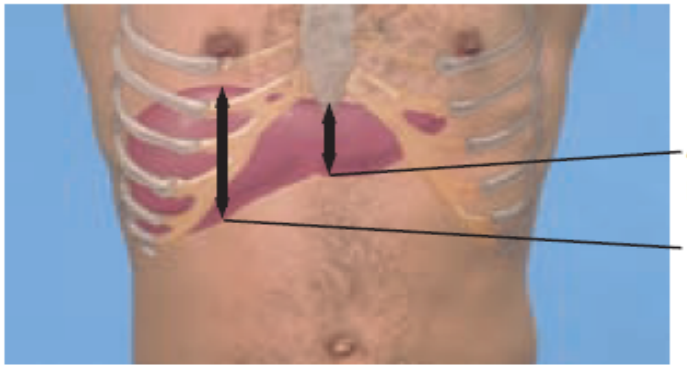
Tumors, a distended viscus



THE LIVER

Percuss span of liver dullness in the midclavicular line (MCL).

Hepatomegaly



Feel the liver edge, if possible, as patient breathes in.

Firm edge of cirrhosis

Measure its distance from the costal margin in the MCL.

Increased in hepatomegaly—may be missed (as below) by starting palpation too high in the RUQ

EXAMINATION TECHNIQUES

POSSIBLE FINDINGS





Note any tenderness or masses.

Tender liver of hepatitis or congestive heart failure; tumor mass

THE SPLEEN

Percuss across left lower anterior chest, noting change from tympany to dullness.

Try to **feel** spleen with the patient

- Supine
- Lying on the right side with legs flexed at hips and knees

Splenomegaly



EXAMINATION TECHNIQUES	POSSIBLE FINDINGS

THE KIDNEYS

Try to **palpate** each kidney.

Enlargement from cysts, cancer, hydronephrosis



Check for costovertebral angle (CVA) tenderness.

Tender in kidney infection



THE AORTA

- **Palpate** the aorta's pulsations. In older people, **estimate** its width.



Periumbilical mass with expansile pulsations ≥3 cm in diameter in abdominal aortic aneurysm. Assess further due to risk of rupture.

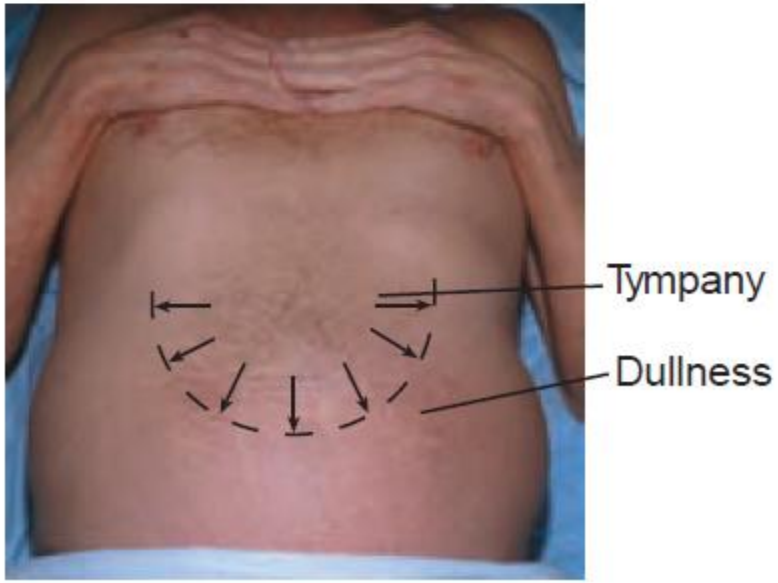
EXAMINATION TECHNIQUES

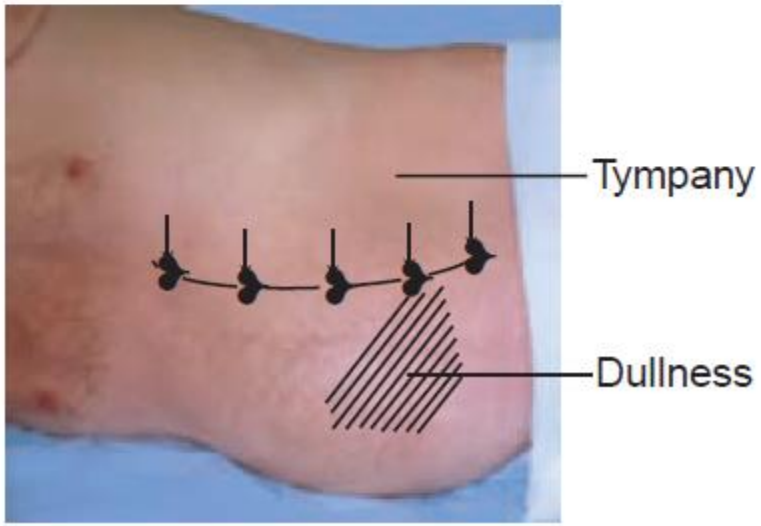
POSSIBLE FINDINGS

ASSESSING ASCITES

Palpate for shifting dullness. Map areas of tympany and dullness with patient supine, then lying on side (see below).

Ascitic fluid usually shifts to dependent side, changing the margin of dullness (see below)



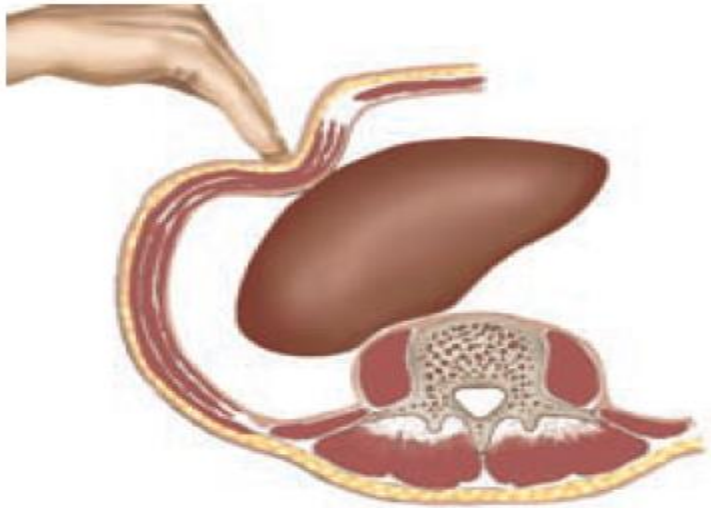


- Check for a fluid wave. Ask patient or an assistant to press edges of both hands into midline of abdomen. Tap one side and feel for a wave transmitted to the other side.



A palpable wave suggests but does not prove ascites.

EXAMINATION TECHNIQUES	POSSIBLE FINDINGS
Ballotte an organ or mass in an ascitic abdomen. Place your stiffened and straightened fingers on the abdomen, briefly jab them toward the structure, and try to touch its surface.	Your hand, quickly displacing the fluid, stops abruptly as it touches the solid surface.



ASSESSING POSSIBLE APPENDICITIS

Ask:	In *classic appendicitis:*
"Where did the pain begin?"	Near the umbilicus
"Where is it now?"	Right lower quadrant (RLQ)
Ask patient to cough. "Where does it hurt?"	RLQ

Palpate for local tenderness.	RLQ tenderness
Palpate for muscular rigidity.	RLQ rigidity
Perform a rectal examination and, in women, a pelvic examination (see Chaps. 14 and 15).	Local tenderness, especially if appendix is retrocecal
• *Rovsing's sign:* Press deeply and evenly in the *left* lower quadrant. Then quickly withdraw your fingers.	Pain in the *right* lower quadrant during *left*-sided pressure suggests appendicitis (a *positive* Rovsing's sign).
• *Psoas sign:* Place your hand just above the patient's right knee. Ask the patient to raise that thigh against your hand. Or, ask the	Pain from irritation of the psoas muscle suggests an inflamed appendix (a *positive* psoas sign).

EXAMINATION TECHNIQUES	POSSIBLE FINDINGS
patient to turn onto the left side. Then extend the patient's right leg at the hip to stretch the psoas muscle.	
• *Obturator sign:* Flex the patient's right thigh at the hip, with the knee bent, and rotate the leg internally at the hip, which stretches the internal obturator muscle.	Right hypogastric pain in a *positive* obturator sign, suggesting irritation of the obturator muscle by an inflamed appendix.

ASSESSING POSSIBLE ACUTE CHOLECYSTITIS

Auscultate, percuss, and palpate the abdomen for tenderness.	Bowel sounds may be active or decreased; tympany may increase with an ileus; RUQ tenderness.
Assess for *Murphy's sign*. Hook your thumb under the right costal margin at edge of rectus muscle, and ask patient to take a deep breath.	Sharp tenderness and a sudden stop in inspiratory effort constitute a *positive* Murphy's sign.

RECORDING YOUR FINDINGS

Recording the Physical Examination—The Abdomen

"Abdomen is protuberant with active bowel sounds. It is soft and nontender; no masses or hepatosplenomegaly. Liver span is 7 cm and in the right MCL; edge is smooth and palpable 1 cm below the right costal margin. Spleen and kidneys not felt. No CVA tenderness."

OR

"Abdomen is flat. No bowel sounds heard. It is firm and boardlike, with increased tenderness, guarding, and rebound in the right midquadrant. Liver percusses to 7 cm in the MCL; edge not felt. Spleen and kidneys not felt. No palpable mass. No. CVA tenderness." *(Suggests peritonitis from possible appendicitis; see pp. 207–208.)*

Thank you

